Notes: Pesendorfer and Schmidt-Dengler (ReStud, 2008)

Asymptotic Least Squares Estimators for Dynamic Games

Contents

1 Big picture

- **Research question.** How can researchers identify and estimate the structural primitives of dynamic games with finite actions when equilibrium behavior is observed over time?
- **Main idea.** A Markov perfect equilibrium implies a system of equilibrium equations linking observed conditional choice probabilities, transition probabilities, and payoff parameters. The paper uses this equation system as the foundation for identification and estimation.
- **Empirical object.** The main empirical inputs are first-stage estimates of players' conditional choice probabilities and state transition probabilities. The second stage chooses structural parameters so that the model-implied equilibrium choice probabilities match the estimated ones as closely as possible.
- **Estimator class.** The paper defines a broad class of **asymptotic least squares** estimators. These estimators minimize a weighted quadratic distance between estimated choice probabilities and equilibrium-implied choice probabilities.
- **Unifying insight.** Several well-known two-step estimators, including Hotz-Miller moment estimators and Aguirregabiria-Mira pseudo-maximum likelihood estimators, are members of the same asymptotic least squares class. They differ mainly in the weights they assign to individual equilibrium conditions.
- **Identification result.** Dynamic games are generally underidentified without restrictions. Even if the discount factor and shock distribution are known, only a limited number of payoff parameters can be identified from the equilibrium choice-probability equations.
- **Efficiency result.** The paper derives the asymptotically efficient weight matrix. The efficient estimator accounts for both the sampling error in auxiliary estimates and how the equilibrium equations change with those auxiliary estimates.
- **Monte Carlo result.** The efficient estimator performs best for moderate and large samples. In very small samples, simpler estimators such as PML or identity-weight least squares can be competitive because the efficient weight matrix is itself estimated noisily.
- **Economic takeaway.** The paper turns dynamic-game estimation into a transparent two-step problem: estimate behavior first, then fit structural payoffs through equilibrium restrictions.

2 Incremental contribution of the paper

- **Extends two-step dynamic discrete-choice methods to games.** Hotz and Miller style estimators were developed for single-agent dynamic problems. Pesendorfer and Schmidt-Dengler show how a similar logic applies to multi-player dynamic games.

- **Uses equilibrium equations as the organizing object.** Instead of solving a nested fixed point at every parameter vector, the paper begins from the equation system that characterizes Markovian equilibrium choice probabilities.
- **Clarifies equilibrium multiplicity.** Cross-sectional estimation of games is complicated because different markets may play different equilibria. The paper emphasizes time-series data generated by a single path of play, which avoids cross-sectional equilibrium-selection assumptions when behavior is Markovian.
- **Provides identification conditions.** The paper explains why dynamic games are underidentified and gives rank conditions under which payoff parameters are exactly identified after imposing restrictions.
- **Unifies existing estimators.** Moment estimators, PML estimators, iterated PML estimators, and least-squares estimators can all be expressed as asymptotic least squares estimators with different weight matrices.
- **Derives the efficient weight matrix.** The paper shows how to combine equilibrium conditions optimally in large samples.
- **Computational message.** Two-step estimators can be simpler than full maximum likelihood because they avoid repeatedly solving dynamic games at every candidate parameter vector.
- **Practical comparison.** A Monte Carlo exercise compares efficient least squares, identity-weight least squares, PML, and iterated PML across sample sizes and equilibria.

3 The motivating example

3.1 Static two-player entry game

The paper first illustrates the estimation logic in a static two-player, two-action game. Each firm chooses whether to be active or not active. Demand has three public states, and each firm has a privately observed profitability shock.

The payoff from being active depends on demand and on whether the rival is also active. Monopoly profits exceed duopoly profits. The private shock enters additively and is drawn from a known distribution. Because shocks have full support, every action is chosen with positive probability in every state.

3.2 Beliefs and equilibrium choice probabilities

Let $\sigma_i(s)$ be firm i's belief that the rival is active in state s. Given beliefs, the best-response condition can be integrated over the private shock distribution to obtain a choice probability:

$$p_i(s) = \Psi_i(\sigma_i(s), \theta),$$

where θ contains payoff parameters.

In equilibrium, beliefs must be consistent with the rival's behavior. This yields a fixed-point equation in choice probabilities:

$$p = \Psi(p, \theta).$$

This equation is the core object of the paper. It relates the observable equilibrium choice probabilities to structural payoff parameters.

3.3 Asymptotic least squares in the example

Suppose the econometrician observes a long time series of states and actions. The first step estimates the conditional choice probabilities $\hat{p}$ from sample frequencies. The second step chooses θ to make

$$\hat{p} - \Psi(\hat{p}, \theta)$$

close to zero. For a weight matrix W, the estimator solves

$$\min_{\theta} [\hat{p} - \Psi(\hat{p}, \theta)]' W [\hat{p} - \Psi(\hat{p}, \theta)].$$

Interpretation. Different choices of W generate different asymptotic least squares estimators. The model is estimated by fitting the equilibrium equations, not by solving the game repeatedly inside a likelihood.

4 Dynamic game model

4.1 States, actions, and shocks

The general model is a discrete-time dynamic game with a fixed set of players $i = 1, \dots, N$.

- Each player has a public state variable $s_i \in S_i$.
- The vector of all public states is $s = (s_1, \dots, s_N) \in S$.
- Each player chooses an action $a_i \in A_i = \{0, 1, \dots, K\}$.
- Actions are chosen simultaneously after players observe the public state and their own private payoff shocks.
- The econometrician and other players do not observe a player's private shocks.
- The private shocks are conditionally independent over time and across players, given the relevant state.

The conditional independence of private shocks keeps the payoff-relevant state space small. If shocks were serially correlated, players would have to form beliefs about rivals' private states from past actions, greatly increasing the dimensionality of the problem.

4.2 Payoffs and transitions

The state transition law is

$$g(a, s, s') = \Pr(s_{t+1} = s' \mid a_t = a, s_t = s),$$

where $a = (a_1, \ldots, a_N)$ is the joint action profile.

Player i's period payoff has two parts:

- an action-state payoff $\pi_i(a, s)$; and
- an action-specific private shock.

Players discount future payoffs by a common discount factor $\beta < 1$.

4.3 Markov perfect equilibrium

The equilibrium concept is Markov perfect equilibrium. Each player's strategy maps the public state and the player's own shock into an action. Beliefs are Markovian and must be consistent with the strategies.

The ex ante value function V_i is defined before the private shocks are observed. In matrix form, the paper writes the value function associated with beliefs σ_i as

$$V_i(\sigma_i) = [I - \beta\sigma_i G]^{-1}\Big[\sigma_i \Pi_i + D_i(\sigma_i)\Big],$$

where Π_i is the vector of period payoffs, G is the transition matrix, and $D_i(\sigma_i)$ collects expected payoff-shock terms.

4.4 Equilibrium equation system

For each player, action, and state, the paper derives best-response probabilities by comparing continuation values across actions and integrating over private shocks.

The resulting equilibrium condition can be written compactly as

$$h(p, g, \theta) = p - \Psi(p, g, \theta) = 0.$$

Here:

- p is the vector of equilibrium conditional choice probabilities;
- g is the state transition law;
- θ collects payoff, transition, shock-distribution, and discount-factor parameters; and
- $\Psi(p, g, \theta)$ is the vector of model-implied best-response probabilities.

Economic message: the equilibrium conditions create a bridge from observable behavior to payoff primitives.

4.5 Existence and special cases

The paper shows that the equilibrium equation system has a solution, and hence a Markov perfect equilibrium exists. It also discusses two important special cases.

- **Symmetric games.** When payoffs, transitions, and strategies are symmetric, the number of equilibrium equations falls, easing computation.
- **Static and single-agent models.** Static games correspond to the special case $\beta = 0$. Single-agent dynamic models correspond to $N = 1$.

5 Identification

5.1 Why time-series data help

The paper focuses on time-series data from a single path of play. This is important because dynamic games can have multiple equilibria.

With a single Markovian time series, the same equilibrium governs behavior whenever the same payoff-relevant state recurs. This avoids the need to assume that different markets or cross-sectional units all select the same equilibrium.

The econometrician is assumed to observe enough variation to estimate:

- conditional choice probabilities $p(a \mid s)$; and
- transition probabilities $g(a, s, s')$.

5.2 Underidentification

The key identification result is that not all primitives of a dynamic game are identified without restrictions.

Even when the shock distribution F and the discount factor β are fixed, at most

$$Km_sN$$

payoff parameters can be identified, where m_s is the number of public states. But the unrestricted payoff vector contains many more parameters, because payoffs can depend on each action profile and state.

Economic message: the degree of underidentification grows with the number of players. The number of potential payoff terms grows exponentially with N, while the number of equilibrium choice-probability equations grows only linearly with N.

5.3 Restrictions and rank condition

To achieve identification, the researcher imposes restrictions on the payoff vector. The paper writes these as linear restrictions:

$$R_i \Pi_i = r_i.$$

The equilibrium conditions can then be written as a linear system in payoffs:

$$X_i(p, g, \beta)\Pi_i + Y_i(p, g, \beta) = 0.$$

Combining equilibrium equations with payoff restrictions yields a square system. Identification reduces to a rank condition: if the augmented matrix has full rank, the payoff vector is exactly identified. Additional restrictions generate overidentification.

5.4 Examples of identifying restrictions

The paper gives restrictions that are common in entry and investment games.

- **No dependence on rivals' states.** A player's payoff can depend on its own state and the joint action profile, but not directly on the state variables of other players.
- **Payoff normalization for action 0.** The payoff from a baseline action, such as inactivity, is fixed exogenously. This is needed because choice behavior identifies payoff differences rather than absolute payoff levels.

Interpretation. The rank condition tells the researcher whether the imposed economic restrictions are strong enough to recover the payoff parameters.

6 Estimation

6.1 Two-step asymptotic least squares

The estimator has two steps.

- **First step.** Estimate auxiliary objects: conditional choice probabilities $\hat{p}$ and transition probabilities $\hat{g}$.
- **Second step.** Choose structural parameters θ to minimize the weighted distance between estimated choice probabilities and equilibrium-implied choice probabilities.

The second-stage problem is

$$\hat{\theta}(W_T) = \arg\min_{\theta} \left[\hat{p} - \Psi(\hat{p}, \hat{g}, \theta)\right]' W_T \left[\hat{p} - \Psi(\hat{p}, \hat{g}, \theta)\right].$$

Under regularity and identification assumptions, the asymptotic least squares estimator is consistent and asymptotically normal.

6.2 Auxiliary estimates

When states and actions are observed frequently, the auxiliary probabilities can be estimated by sample frequencies:

$$\hat{p}(k,i,s) = \frac{n_{kis}}{n_{is}}, \qquad \hat{g}(a,s,s') = \frac{n_{ass'}}{\sum_{s''} n_{ass''}}.$$

Other first-stage methods, such as maximum likelihood or kernel smoothing, can also be used depending on the data structure.

6.3 Efficient weight matrix

Because the equilibrium equation system can be overidentified, the choice of weights matters. The efficient weight matrix accounts for:

- the covariance matrix of the auxiliary estimates; and
- the derivative of the equilibrium equations with respect to the auxiliary parameters p and g.

Conceptually, the efficient estimator gives less weight to noisy equilibrium conditions and accounts for the fact that errors in first-stage choice probabilities propagate through $\Psi(p,g,\theta)$.

The paper's efficient matrix has the form

$$W_0 = \left(A\Sigma A'\right)^{-1},$$

where Σ is the asymptotic covariance matrix of the auxiliary estimates and A is the derivative adjustment generated by the equilibrium equations.

6.4 Moment estimators as asymptotic least squares

Hotz-Miller style moment estimators use moment conditions based on observed choices minus model-implied choice probabilities, interacted with instruments. Pesendorfer and Schmidt-Dengler show that these moment estimators can be rewritten as asymptotic least squares estimators.

The important implication is that the choice of instruments and GMM weight matrix determines the implicit weight placed on each equilibrium condition. Common instruments such as current and lagged state variables can average over equilibrium equations in ways that are not efficient.

To achieve efficiency, the moment conditions should effectively include separate conditions for each player-state-action equation, with the weight matrix chosen as the efficient asymptotic least squares matrix.

6.5 Pseudo-maximum likelihood as asymptotic least squares

The pseudo-maximum likelihood estimator sets observed choice frequencies equal to the pseudo-likelihood choice probabilities. The paper shows that its first-order condition corresponds to an asymptotic least squares estimator with weights related to the inverse covariance matrix of choice probabilities.

This weighting is efficient in some single-agent settings because a derivative term vanishes at the optimum. In multi-agent games, that derivative need not vanish: player j's choices do not maximize player i's continuation value. Thus PML generally fails to use the equilibrium conditions efficiently in dynamic games.

6.6 Iterated PML

Aguirregabiria and Mira's iterated PML updates the auxiliary choice probabilities using the model-implied choice probabilities and repeats the PML step. The paper includes this estimator in the Monte Carlo comparison.

Interpretation. Iteration can help in some cases, but the Monte Carlo evidence shows that it can also create substantial bias in some equilibria.

7 Monte Carlo design and findings

7.1 Design

The Monte Carlo model is intentionally simple.

- There are two players.
- Each player has two actions, 0 and 1.
- Each player has two states, 0 and 1.
- Private profitability shocks are standard normal.
- The discount factor is 0.9.
- The structural parameters estimated are entry/switching cost c, monopoly payoff π^1, and duopoly payoff π^2.
- The true values are $c = -0.2$, $\pi^1 = 1.2$, and $\pi^2 = -1.2$.

The game has multiple equilibria. The paper studies three equilibria: two asymmetric equilibria and one symmetric equilibrium.

For each equilibrium, the authors simulate samples of length $T = 100$, 1,000, 10,000, and 100,000, with 1,000 repetitions for each design. They compare four estimators:

- LS-E: efficient asymptotic least squares;
- LS-I: identity-weight asymptotic least squares;
- PML: pseudo-maximum likelihood; and
- k-PML: 20-step iterated pseudo-maximum likelihood.

7.2 Main findings

- **LS-E is best for moderate and large samples.** It is selected by the MSE criterion in eight of the twelve Monte Carlo specifications.
- **Small samples are different.** For $T = 100$, LS-E can perform poorly because the optimal weight matrix is estimated imprecisely.
- **PML is useful in small samples.** PML often performs better than LS-E at $T = 100$, but it is generally worse in larger samples.
- **Identity weights are simple but not optimal.** LS-I is never selected as the preferred estimator by the MSE criterion.
- **k-PML can be biased.** Iterated PML performs reasonably well in equilibrium (i), but exhibits large bias in equilibria (ii) and (iii), especially as the number of iterations increases.
- **Computation is mostly about equilibrium solving.** In the CPU-time table, solving the model can be much more time-consuming than estimating the parameters, especially as the number of firms and demand states increases.

8 Identification and interpretation

8.1 What the paper identifies

The paper identifies payoff parameters from repeated observations of equilibrium behavior under a maintained Markov perfect equilibrium and maintained restrictions on payoffs, shocks, transitions, and discounting.

The key empirical assumption is that the estimated conditional choice probabilities are equilibrium probabilities generated by one Markovian equilibrium.

8.2 Multiplicity concern

Multiple equilibria are not a nuisance only for theory; they directly affect estimation. In cross-sectional data, different markets may play different equilibria. In a single time series, the Markovian assumption implies that the same equilibrium governs repeated visits to the same state. This is why the paper's time-series approach is attractive.

8.3 Auxiliary-estimation concern

Two-step estimators depend on accurate first-stage estimates. If some states are rarely visited, choice probabilities and transition probabilities are estimated imprecisely. This affects the second stage and can make the efficient weight matrix unstable in small samples.

8.4 Interpretation of efficiency

Efficiency here means asymptotic efficiency within the asymptotic least squares class. It does not eliminate model misspecification risk. If the equilibrium concept, payoff restrictions, shock distribution, or transition law are wrong, the estimator targets the pseudo-true parameter implied by the wrong restrictions.

9 Tables

9.1 Tables 1 and 2: Monte Carlo results for equilibria (i) and (ii)

Read Table 1:

- Table 1 reports estimates for asymmetric equilibrium (i).
- The true values are $c = -0.2$, $\pi^1 = 1.2$, and $\pi^2 = -1.2$.
- At $T = 100$, k-PML has the lowest MSE, while LS-E has high MSE because the efficient weight matrix is noisy.
- At $T = 1{,}000$ and above, LS-E has the lowest MSE and becomes highly accurate.

Read Table 2:

- Table 2 reports estimates for asymmetric equilibrium (ii).
- k-PML is strongly biased in this equilibrium, especially for c and π^2.
- LS-E becomes the best estimator once $T \geq 1{,}000$.
- PML and LS-I are close to LS-E in some larger samples but do not dominate it.

Economic message: the efficient weight matrix matters most when the sample is large enough to estimate it well. Iteration is not automatically an improvement in games with multiple equilibria.

TABLE 1

Monte Carlo results, equilibrium (i)

T	Estimator	c		π^1		π^2		MSE
100	LS-E	−0·116	(0·768)	1·424	(1·472)	−1·367	(1·108)	406·617
	LS-I	−0·292	(0·400)	1·086	(0·493)	−1·052	(0·511)	70·732
	PML	−0·260	(0·310)	1·083	(0·344)	−1·067	(0·372)	38·738
	k-PML	−0·231	(0·163)	1·225	(0·309)	−1·186	(0·229)	17·655
1000	LS-E	−0·195	(0·030)	1·207	(0·077)	−1·208	(0·067)	1·156
	LS-I	−0·213	(0·104)	1·185	(0·093)	−1·189	(0·134)	3·791
	PML	−0·209	(0·111)	1·187	(0·107)	−1·191	(0·129)	4·061
	k-PML	−0·204	(0·040)	1·197	(0·090)	−1·202	(0·061)	1·334
10,000	LS-E	−0·200	(0·008)	1·199	(0·022)	−1·200	(0·018)	0·089
	LS-I	−0·203	(0·032)	1·197	(0·028)	−1·196	(0·042)	0·363
	PML	−0·202	(0·034)	1·196	(0·034)	−1·197	(0·040)	0·396
	k-PML	−0·201	(0·012)	1·199	(0·029)	−1·200	(0·018)	0·132
100,000	LS-E	−0·200	(0·003)	1·200	(0·007)	−1·200	(0·005)	0·009
	LS-I	−0·201	(0·010)	1·200	(0·009)	−1·200	(0·013)	0·036
	PML	−0·201	(0·011)	1·200	(0·011)	−1·199	(0·013)	0·042
	k-PML	−0·200	(0·004)	1·200	(0·009)	−1·200	(0·006)	0·013

Table 1: Monte Carlo results, equilibrium (i).

TABLE 2

Monte Carlo results, equilibrium (ii)

T	Estimator	c		π^1		π^2		MSE
100	LS-E	−0·391	(0·559)	1·047	(0·415)	−0·967	(0·577)	93·139
	LS-I	−0·235	(0·509)	1·094	(0·480)	−1·025	(0·678)	99·243
	PML	−0·322	(0·448)	0·954	(0·399)	−0·907	(0·573)	84·802
	k-PML	−0·499	(0·193)	1·031	(0·245)	−0·734	(0·186)	46·639
1000	LS-E	−0·219	(0·153)	1·181	(0·126)	−1·169	(0·192)	7·759
	LS-I	−0·208	(0·173)	1·191	(0·140)	−1·176	(0·207)	9·319
	PML	−0·222	(0·165)	1·165	(0·139)	−1·151	(0·207)	9·377
	k-PML	−0·491	(0·060)	0·998	(0·065)	−0·749	(0·052)	33·937
10,000	LS-E	−0·198	(0·021)	1·199	(0·025)	−1·202	(0·030)	0·192
	LS-I	−0·202	(0·056)	1·198	(0·046)	−1·197	(0·066)	0·959
	PML	−0·205	(0·054)	1·193	(0·045)	−1·192	(0·067)	0·953
	k-PML	−0·493	(0·029)	0·991	(0·027)	−0·750	(0·036)	33·535
100,000	LS-E	−0·200	(0·002)	1·200	(0·007)	−1·200	(0·005)	0·008
	LS-I	−0·201	(0·017)	1·200	(0·014)	−1·199	(0·020)	0·088
	PML	−0·201	(0·016)	1·199	(0·014)	−1·199	(0·020)	0·087
	k-PML	−0·491	(0·032)	0·991	(0·025)	−0·752	(0·047)	33·245

Table 2: Monte Carlo results, equilibrium (ii).

9.2 Tables 3 and 4: Symmetric equilibrium and computation times

Read Table 3:

- Table 3 reports estimates for the symmetric equilibrium (iii).
- At $T = 100$, PML and k-PML have lower MSE than LS-E.
- For $T = 10{,}000$ and $T = 100{,}000$, LS-E has the lowest MSE.
- In this symmetric equilibrium, LS-E, LS-I, and PML become quite close in moderate and large samples.

Read Table 4:

- Table 4 reports CPU times for solving a Cournot model and estimating parameters.
- Solving the model becomes very costly as the number of firms and demand states rises.
- Estimation is generally faster than solving for equilibrium, but LS-E can become slower than PML because it requires construction of the efficient weight matrix.
- For large state spaces, computational feasibility may favor simpler weight matrices even if they are not asymptotically efficient.

Economic message: asymptotic efficiency and computational tractability are not the same criterion. The efficient estimator is attractive statistically, but in large games the cost of computing the weight matrix can matter.

TABLE 3

Monte Carlo results, equilibrium (iii)

T	Estimator	c		π^1		π^2		MSE
100	LS-E	−0·439	(0·794)	1·036	(0·596)	−0·928	(0·963)	206·942
	LS-I	−0·240	(0·491)	1·101	(0·476)	−1·008	(0·627)	90·878
	PML	−0·326	(0·433)	0·964	(0·393)	−0·896	(0·541)	79·729
	k-PML	−0·504	(0·190)	1·037	(0·229)	−0·722	(0·169)	46·474
1000	LS-E	−0·219	(0·179)	1·181	(0·145)	−1·173	(0·212)	9·925
	LS-I	−0·209	(0·180)	1·189	(0·147)	−1·179	(0·211)	9·934
	PML	−0·227	(0·170)	1·156	(0·141)	−1·147	(0·207)	9·733
	k-PML	−0·511	(0·062)	0·992	(0·066)	−0·734	(0·052)	36·835
10,000	LS-E	−0·201	(0·053)	1·198	(0·042)	−1·200	(0·063)	0·858
	LS-I	−0·200	(0·057)	1·199	(0·046)	−1·201	(0·065)	0·952
	PML	−0·202	(0·053)	1·195	(0·043)	−1·197	(0·064)	0·883
	k-PML	−0·510	(0·025)	0·989	(0·022)	−0·733	(0·030)	36·082
100,000	LS-E	−0·201	(0·017)	1·199	(0·013)	−1·198	(0·020)	0·085
	LS-I	−0·200	(0·018)	1·200	(0·014)	−1·199	(0·021)	0·096
	PML	−0·201	(0·017)	1·199	(0·014)	−1·199	(0·021)	0·091
	k-PML	−0·508	(0·031)	0·990	(0·020)	−0·736	(0·045)	35·700

Table 3: Monte Carlo results, equilibrium (iii).

TABLE 4

CPU times

	Model parameters							
N	2	2	5	5	10	10	15	15
Demand levels	2	10	2	10	2	10	2	10
	CPU times							
Solve model	0·98	2·70	0·66	34·46	5·24	582·70	42·31	4720·82
Estimate $\hat{\boldsymbol{p}}$	1·12	1·12	1·12	1·19	1·13	1·62	1·21	3·52
	(0·03)	(0·01)	(0·01)	(0·00)	(0·00)	(0·01)	(0·01)	(0·77)
LS-I	0·14	0·04	0·04	0·06	0·02	0·04	0·03	0·10
	(0·19)	(0·00)	(0·01)	(0·01)	(0·00)	(0·01)	(0·00)	(0·02)
LS-E	0·05	0·21	0·08	3·47	0·73	48·21	3·42	288·65
	(0·02)	(0·03)	(0·01)	(0·01)	(0·02)	(0·74)	(0·02)	(48·92)
PML	0·04	0·03	0·05	0·08	0·02	0·11	0·03	0·20
	(0·01)	(0·01)	(0·01)	(0·01)	(0·00)	(0·02)	(0·01)	(0·03)
k-PML	0·86	0·67	0·62	1·76	0·88	9·97	1·91	42·75
	(0·01)	(0·04)	(0·01)	(0·01)	(0·02)	(0·11)	(0·02)	(9·77)

Note: For each set of model parameters, the model was solved once and simulated five times. Reported central processing unit (CPU) times are averages in seconds. S.E. are in parentheses.

Table 4: CPU times.

10 Short summary

- The paper studies structural estimation of dynamic games with finite actions.
- It derives a system of equilibrium equations for Markov perfect equilibria.
- These equations link observed conditional choice probabilities, transition probabilities, and structural payoff parameters.
- The model is generally underidentified without restrictions, even if the discount factor and shock distribution are known.
- The paper gives rank conditions for identification once restrictions are imposed.
- Estimation is two-step: first estimate choice and transition probabilities; then fit the equilibrium equations.
- Asymptotic least squares estimators minimize a weighted distance between estimated and model-implied equilibrium choice probabilities.
- Many existing estimators are asymptotic least squares estimators with different weight matrices.
- The efficient weight matrix accounts for both first-stage sampling variation and the derivative of the equilibrium equations with respect to auxiliary estimates.
- PML is generally not efficient in multi-agent dynamic games because strategic interactions make derivative terms nonzero.
- The Monte Carlo study finds that efficient least squares is best for moderate and large samples.
- In small samples, simpler estimators can perform better because the optimal weight matrix is noisy.
- Iterated PML can be severely biased in some equilibria.

11 Conclusion

11.1 Core conclusion

Pesendorfer and Schmidt-Dengler show that dynamic-game estimation can be organized around equilibrium choice-probability equations. These equations provide both identification insights and a practical two-step estimation method.

11.2 Economic intuition

Players' observed choices reveal their equilibrium beliefs and continuation values. Once conditional choice probabilities and transition probabilities are estimated, structural payoff parameters must rationalize the observed behavior through the best-response equations of the game.

11.3 Incremental contribution revisited

The paper's main contribution is not a new dynamic game application, but a general estimation framework. It places Hotz-Miller moment estimators, Aguirregabiria-Mira PML, and other dynamic-game estimators inside one asymptotic least squares class.

11.4 Implications

The paper clarifies that estimator choice is largely weight-matrix choice. Applied researchers should pay attention to how their estimator weights equilibrium conditions, whether their instruments preserve state-level conditions, and whether the data are rich enough to estimate the efficient weight matrix reliably.

11.5 Limitations

The framework assumes finite actions, a manageable finite state space, conditionally independent private shocks, and a Markovian equilibrium. It is less suited for games with many players, large state spaces, serially correlated private information, or continuous action spaces unless additional structure is imposed.

11.6 Takeaway

The enduring takeaway is that dynamic games can be estimated by first measuring behavior and then imposing equilibrium logic. The power of the approach comes from separating the nonparametric first-stage estimation of behavior from the structural second-stage fitting of payoffs.